IRT / RTSM User Acceptance Testing

One-page guide · synthetic example · operational data only · human sign-off required

What UAT confirms

Confirm the staging randomization and drug-supply build against the approved user requirements and specifications before real use. Execute positive, negative and boundary cases; retain dated evidence and an attributable execution record. The toolkit provides templates and a readiness screen for adaptation to the study.

The readiness screen

The automated screen clears only when **all three** hold:

- Every **Critical** test passes
- No open Critical/Major defects; closed defects pass retest; failing tests have a logged defect
- Every requirement names existing tests and every test has been run (Pass or Fail)

Snapshot now: **NOT ready - blocked**.

Open Minor defects need review or a documented waiver. A clear screen is not signed approval.

Bundled synthetic snapshot

Test scripts	29
Passed	22
Failed	2
Open defects	3
Requirements traced	97%

Recomputed when this PDF is built; not a live tracker.

Defect severity drives the gate

Critical	Could break the blind, mis-randomize, mis-dose, or corrupt integrity	Blocks
Major	Significant functional error in a key workflow (resupply, wrong kit, missing alert)	Blocks
Minor	Cosmetic / non-functional (label typo, missing site number)	Fix or waive

Where it runs: the systems (the workbook is the paper trail)

Staging IRT (approved test list) → **browser execution** → **dated evidence** → **workbook** (tests, defects, traceability) → **signed package / eTMF**. Failures also go to the **vendor defect tracker**; the vendor promotes a fixed build back to staging for retest. The workbook never connects to the IRT, you transcribe, on purpose.

Anatomy of an executable test (RAND-02)

A good test has a **specific, verifiable** expected result, not "works correctly".

- **Precondition:** approved synthetic list loaded; the required stratum has an unused slot.
- **Steps:** register a synthetic participant, randomize, record kit and confirmation, then inspect the authorized allocation report.
- **Expected:** the next unused entry, arm and kit match the list; the investigator sees a **kit ID only**.
- **Verify:** every stratum and arm, plus permission failures and boundary cases.

The 8-step workflow

1. Derive test cases from URS + specs (positive · negative · boundary)
2. Load the approved test list into staging IRT
3. Execute each case in the browser; be systematic
4. Record result + dated screenshot evidence
5. Log every failure as a defect with a severity
6. Vendor fixes; retest the new build
7. Check actual test links, execution and retest evidence
8. Evaluate the gate; a named human signs it into the eTMF

Run the interactive tool

From the repository root: `Rscript launch_app.R uat --install`

The local app recomputes the screen as tests and defects change. CSV download contains the test table only; retain defects, traceability and signed evidence separately. Adapt and verify the toolkit before study-specific use.